

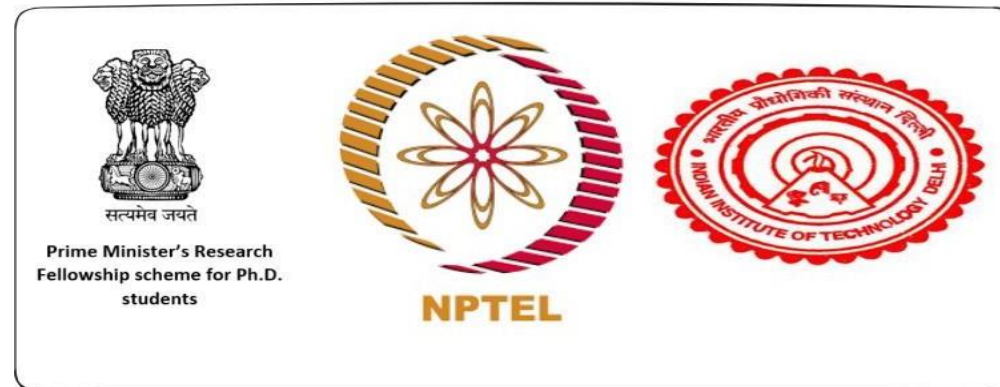
Interaction Session 5

on

Cloud Computing
noc25-cs11

by

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Indian Institute of Technology, Delhi

Points to Remember:

- Total 7 more Discussion sessions to be taught lectures and Tutorials
- Any specific problem, please upload in the discussion forum, I will try to reply within 24 hours
- PPT and Video will be shared with everyone
- Assignments will not be solved but similar problems will be solved.

TOPICS TO BE COVERED

1. SLA Agreement
2. MAP-Reduce
3. SSH
4. Security Groups in EC2

Questions on SLA Agreement

In a SLA negotiation, the provider agreed with the service availability of 98%. The consumer runs the application for X hours/day. At the end of one month [31 days], the total service outage was 12 hrs. However, SLA negotiation (in terms of service availability) is honored.

- Total time for which the application will run (in a month) = $(X * 31)$ hours
- Outage time = 12 hours --> {the amount of time that the network is unavailable during a defined period.}
- Therefore, service duration = $(X * 31) - 12$ hours
- % Availability = $(1 - \text{outage time} / \text{service duration}) * 100 \% = (1 - (12 / ((X * 31) - 12))) * 100$
- % Initial service guarantee = 98% As, final service
- $\text{availability} \geq \text{initial service guarantee}$ $98 \leq (1 - (12 / (X * 31 - 12))) * 100$
- or, $X \geq 19.74$

Questions on Average Demand

Average resource demand is 45 units, Baseline (owned) unit cost is 200 units, Time is 10 hours, Peak resource demand is 100 units. If the cloud is cheaper than owning of computer infrastructures, the utility premium is a. Greater than 2.22

b. Less than 2.22

c. Atleast 4.45

d. Atmost 4.45

A= Average Demand, U= Utility Premium, B=Baseline (owned) unit cost, T=Time, P=PeakDemand, CT=Cloud cost and BT=Owning cost.

Here P=100 units, A=45 units, B=200, T=10.

$U < P/A$

implies $U < 100/45$ or, $U < 2.22$

Utility Pricing in Detail

D(t)	demand for resources $0 < t < T$
P	max (D(t)) : Peak Demand
A	Avg (D(t)) : Average Demand
B	Baseline (owned) unit cost [B_T : Total Baseline Cost]
C	Cloud unit cost [C_T : Total Cloud Cost]
U (=C/B)	Utility Premium [For rental car example, $U=4.5$]

$$C_T = \int_0^T U \times B \times D(t) dt = A \times U \times B \times T$$

$$B_T = P \times B \times T$$

- Because the baseline should handle peak demand

When is cloud cheaper than owning?

$$C_T < B_T \rightarrow A \times U \times B \times T < P \times B \times T$$

$$\rightarrow U < \frac{P}{A}$$

- When utility premium is less than ratio of peak demand to Average demand

MAP Reduce- history

- **Early Inspiration (Before 2003)**
- Functional Programming (Lisp's *map* & *reduce*)
- Distributed Computing Challenges
- **Google's Contribution (2003-2004)**
- 2003: Google internally develops MapReduce (Jeffrey Dean & Sanjay Ghemawat)
- 2004: Research paper published
- **Hadoop and Open-Source Adoption (2005-2008)**
- 2005: Doug Cutting & Mike Cafarella start open-source MapReduce for Apache Nutch
- 2006: Yahoo! launches **Apache Hadoop**
- 2008: Hadoop becomes an Apache Top-Level Project
- **Peak Adoption (2010-2015)**
- Companies like Facebook, Twitter, LinkedIn adopt MapReduce
- 2012: Apache Spark introduces faster in-memory computation
- 2014: Google replaces MapReduce with **Dataflow**
- **Decline & Modern Alternatives (2016-Present)**
- Spark, Flink, and real-time processing replace MapReduce
- Cloud-based solutions dominate big data analytics

MAP Reduce

Mapper

Reducer

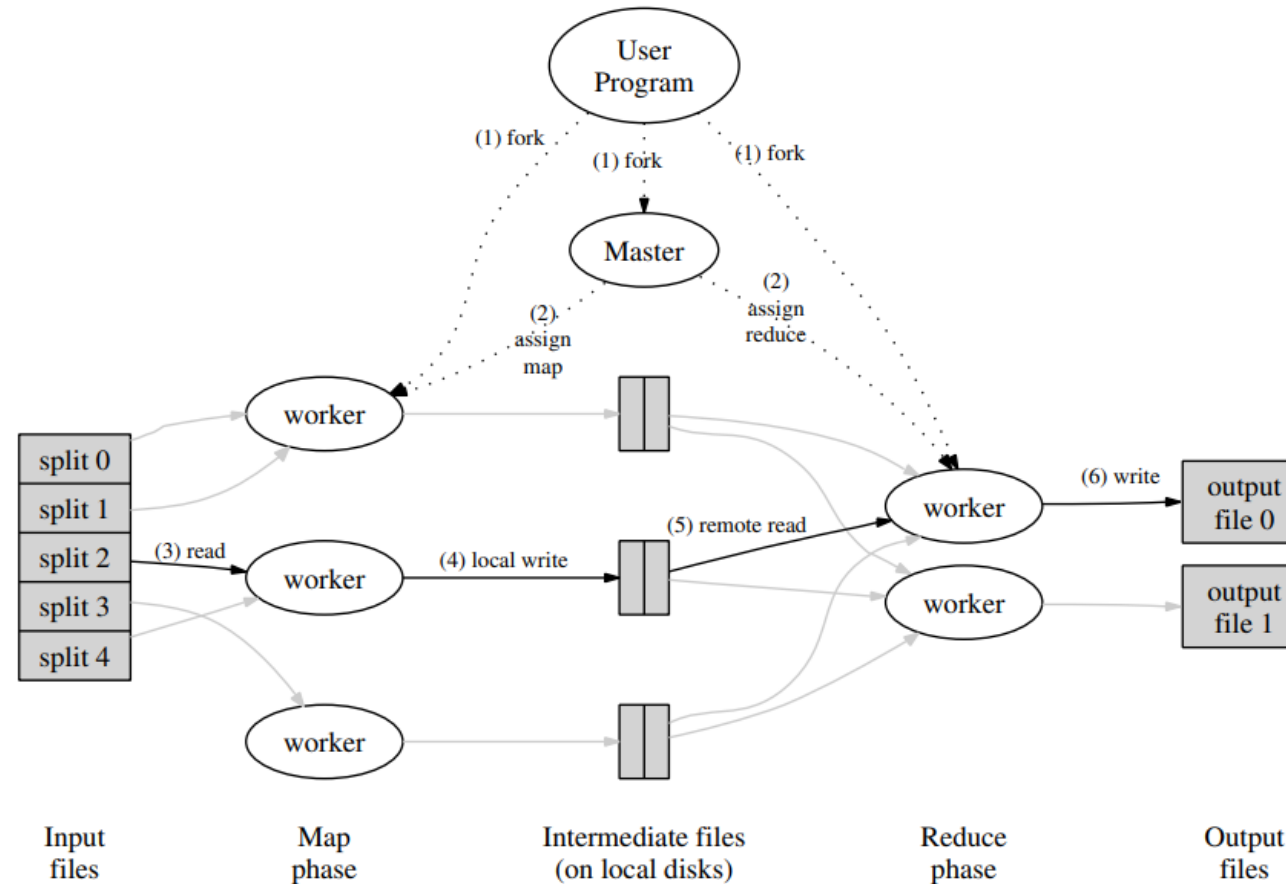
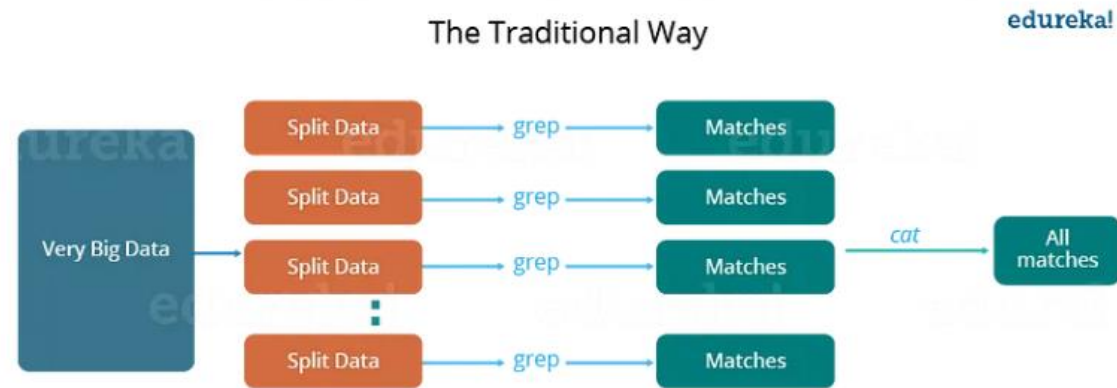
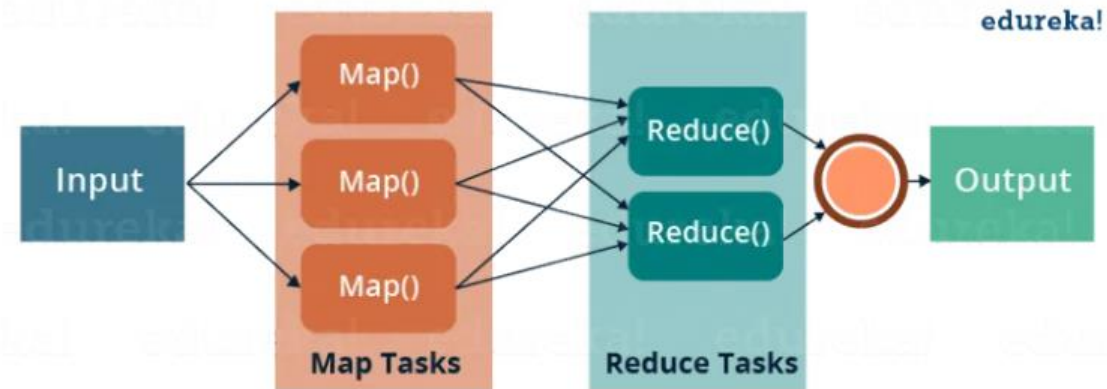


Figure 1: Execution overview

MAP Reduce

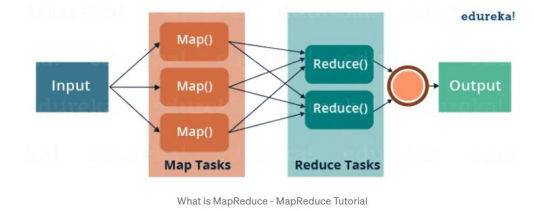
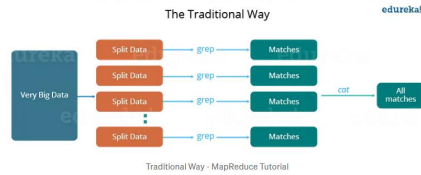


Traditional Way - MapReduce Tutorial

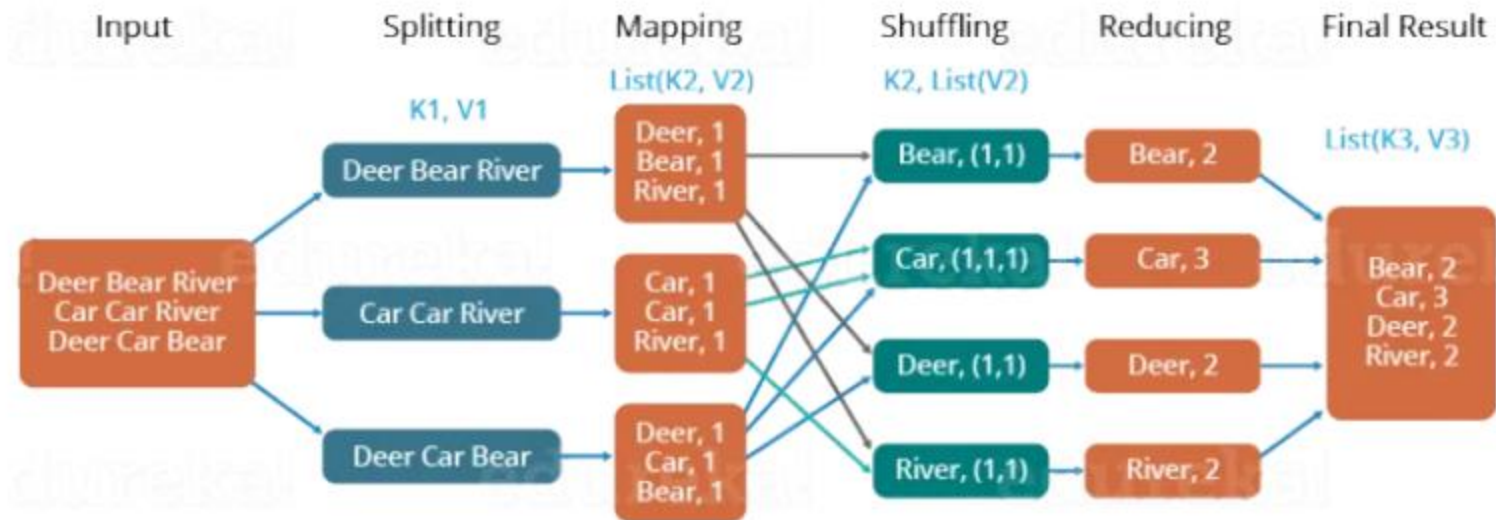


What is MapReduce - MapReduce Tutorial

MAP Reduce



The Overall MapReduce Word Count Process



MapReduce Example - MapReduce Tutorial

MAP Reduce – Case Study

Scenario: Detecting Most Frequent Errors in Server Logs

Imagine you are managing a **large-scale data center** with **thousands of servers**. Each server generates logs containing error messages. You want to **find the most frequent error messages** in real-time across all servers.

How MapReduce Can Solve This?

- **Input: Distributed Server Logs**

- Each server generates a log file like:

```
[ERROR] Disk failure detected  
[INFO] User login successful  
[ERROR] Network timeout  
[ERROR] Disk failure detected
```

```
[WARNING] High CPU usage
```

- These logs are stored **across multiple servers** in a distributed file system.

https://colab.research.google.com/drive/1ZPA_OP1Lb0KON05xweNFIwlhAyv9PGuG?usp=sharing

MAP Reduce

- **Map Phase (Runs on Worker Nodes - Mappers)**
 - Each mapper reads a **portion of logs** from different servers.
 - It extracts **error messages** and emits (error_message, 1) pairs.
 - Example output from a mapper:
("Disk failure detected", 1)
("Network timeout", 1)
("Disk failure detected", 1)
("High CPU usage", 1)

MAP Reduce

- **Shuffle & Sort Phase (Master Node)**
 - The master node **groups identical keys together** and assigns them to reducers.
 - Example:
 - ("Disk failure detected", [1, 1])
 - ("Network timeout", [1])
 - ("High CPU usage", [1])
- **Reduce Phase (Runs on Worker Nodes - Reducers)**
 - Each reducer **sums up** occurrences of each error message.
 - Example output:
 - ("Disk failure detected", 2)
 - ("Network timeout", 1)
 - ("High CPU usage", 1)
- **Final Output: Most Frequent Errors**
 - The final output is stored in a database or visualized in a dashboard, showing the most common errors across all servers.

Question on Map Reduce

In a MapReduce framework the HDFS block size is 64 MB. We have 6 files of size 64KB, 65MB, X MB, Y KB, 67KB and 127MB. 24 blocks are created by Hadoop framework. The size of X and Y are respectively:

The total number of blocks needed for
64 KB, 65 MB, 67 KB, 127 MB is

$$(1+2+1+2)*3=18$$

[3 is the number of replicas]

$$\begin{aligned}\text{Number of remaining blocks} &= (24-18)/3 \\ &= 6/3 = 2 \text{ [3 is the number of replicas]}\end{aligned}$$

X and Y each <64MB,

Hands-on Session Checklist

Step 1

- AWS Free-tier Account

Step 2

- IAM User
- MFA

Step 3

- Budget – zero spend

Step 4

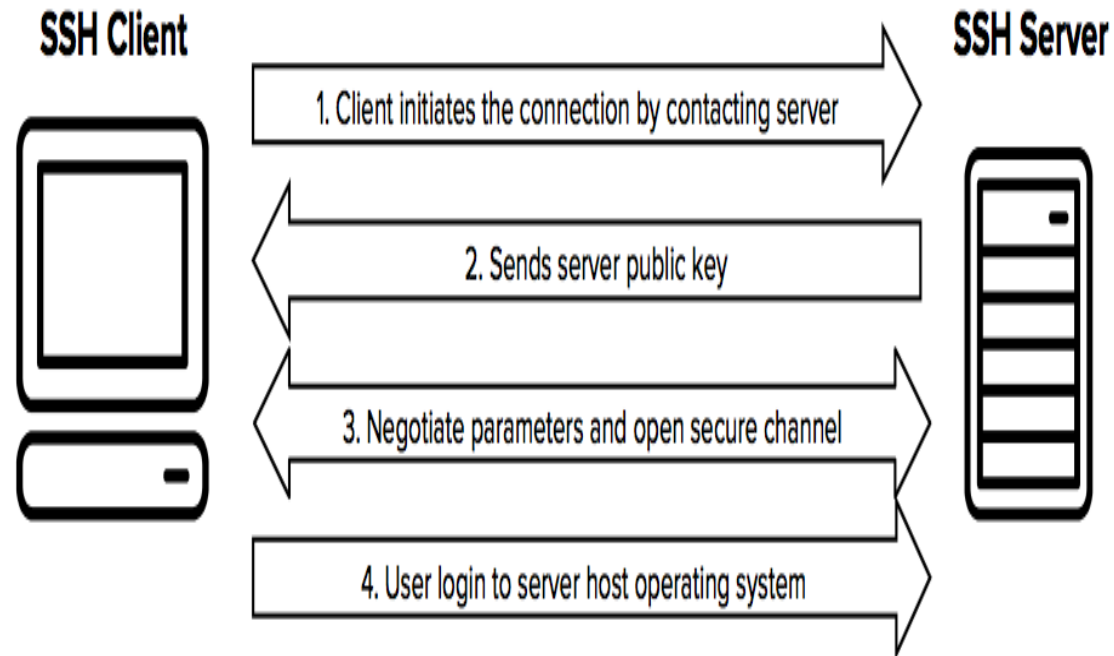
- Create an Instance in EC2

SSH



SSH (Secure Shell) is a cryptographic network protocol used to securely access and manage remote systems over an unsecured network. It encrypts all communication, preventing eavesdropping and tampering. Authentication methods include passwords and public-key cryptography, with the latter being more secure. SSH also supports tunneling, port forwarding, and secure file transfer (SCP/SFTP). It operates on port 22 by default and is essential for remote administration, automation, and secure data transfers.

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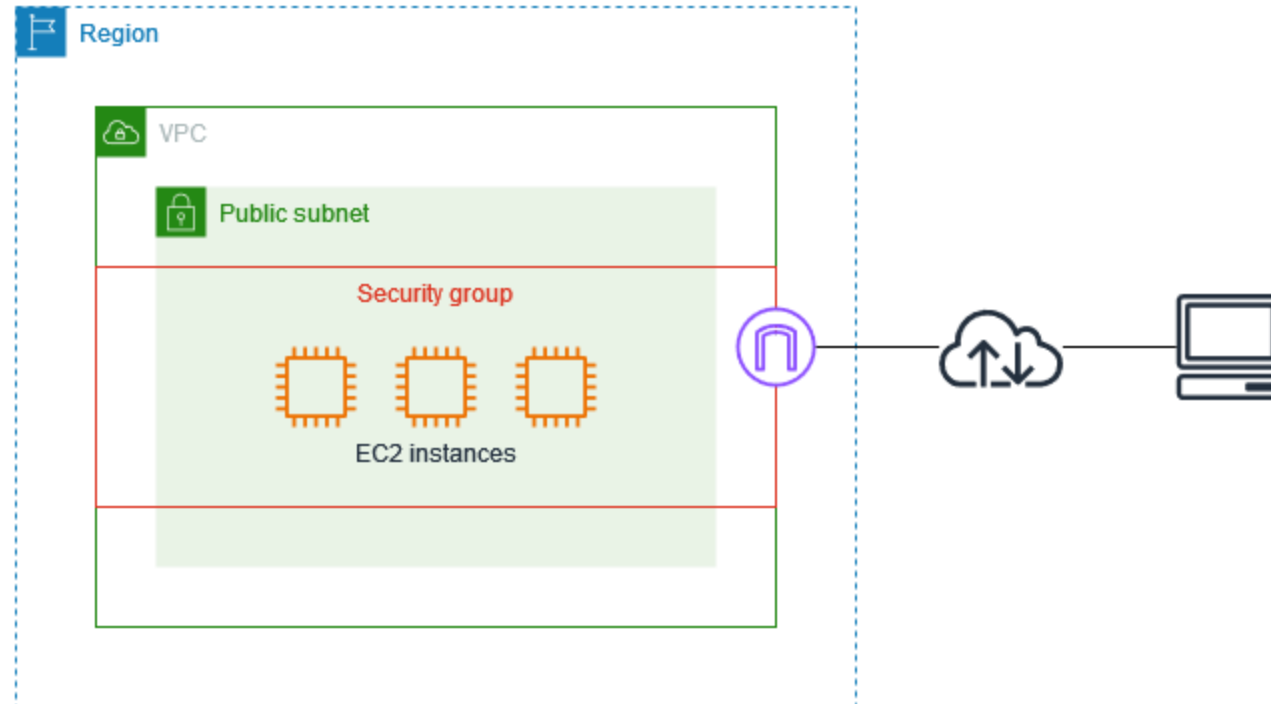
SSH--> EC2

SSH Summary Table

	SSH	Putty	EC2 Instance Connect
Mac	✓		✓
Linux	✓		✓
Windows < 10		✓	✓
Windows >= 10	✓	✓	✓



Security Groups



<https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/creating-security-group.html>

Steps to Connect to EC2 using SSH for windows >10

Step 1: Install an SSH Client (if not already installed)

Windows 11 comes with **OpenSSH Client** pre-installed. To verify:

- Open **Command Prompt** (Win + R → type cmd → Enter).
- Type: ssh
-

If you see SSH usage details, it's installed. Otherwise, install it from:

- **Settings** → Apps → Optional Features → Add OpenSSH Client.

Alternatively, you can use **PuTTY**

Steps to Connect to EC2 using SSH for windows >10

Step 2: Download and Set Up Your EC2 Key Pair

- **Get the .pem Key File:**
 - While launching your EC2 instance, AWS provides a **key pair** (e.g., my-key.pem).
 - Download it and keep it safe.
- **Move the .pem File to a Secure Location:**
 - Example: Move it to C:\Users\YourUser\.ssh\
- **Change File Permissions** (Needed for OpenSSH):
 - Open **PowerShell** as Administrator and run:
`icacls C:\Users\YourUser\.ssh\my-key.pem /inheritance:r`
`icacls C:\Users\YourUser\.ssh\my-key.pem /grant:r %username%:R`

Steps to Connect to EC2 using SSH for windows >10

Step 3: Find Your EC2 Public IP

- Go to **AWS Management Console**.
- Navigate to **EC2 Dashboard** → Instances.
- Find your instance and copy its **Public IPv4 Address** (e.g., 3.25.123.45).

Step 4: Connect Using OpenSSH (Command Prompt/PowerShell)

- Open **Command Prompt** or **PowerShell**.
- Run the SSH command:

```
ssh -i "C:\Users\YourUser\.ssh\my-key.pem" ec2-user@<Your-EC2-Public-IP>
```

```
ssh -i "C:\Users\NPTEL\.ssh\my-key.pem" ec2-user@3.25.123.45
```

COST FOR EC2- public IPV4

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```

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```

<https://aws.plainenglish.io/how-to-connect-to-aws-ec2-instance-using-ssh-and-rdp-ecac5ab6ac77>

<https://aws.plainenglish.io/aws-article-7-integrating-ssh-with-ec2-eecf367a58ca>

In case SSH not working---> do troubleshoot

Connect to your instance i-034466697feb9ef80 (My First Instance) using any of these options

EC2 Instance Connect | Session Manager | SSH client | EC2 Serial Console

Instance ID
i-034466697feb9ef80 (My First Instance)

Public IP address
3.250.26.200

User name

Connect using a custom user name, or use the default user name ec2-user for the AMI used to launch the instance.

Note: In most cases, the guessed user name is correct. However, read your AMI usage instructions to check if the AMI owner has changed the default AMI user name.

Cancel **Connect**

AWS Connect--> Browser based Terminal